

Commerce XI Mathematics & Statistics · Part 1

Unit 1: Sets and Relations

A quick reference based on Unit 1, textbook pages 1–19.

How to use this guide

Use	What to check
Before solving	Identify the universal set, the sets involved, and whether the question asks for elements, a relation, or cardinality.
During solving	Translate every symbol into words, apply the relevant law, and keep complements relative to the stated universal set.
Before answering	Remove repeated elements, check endpoints in intervals, preserve the order in ordered pairs, and verify the required form.

Scope

Representation of sets · intervals · types of sets · set operations · cardinality · ordered pairs · Cartesian products · relations · properties and types of relations.

Textbook Answer Notes

Exercise 1.1, Q11: the empty set $\emptyset$, rather than $\{\emptyset\}$, is an element of the power set. Exercise 1.1, Q12(i): the closed interval $[-3, 0]$ is $-3 \leq x \leq 0$.

1. Set Symbols and How to Read Them

Symbol / notation	Read as	Meaning / use	Example
$x \in A$	x belongs to A	x is an element of set A.	$2 \in \{1, 2, 3\}$
$x \notin A$	x does not belong to A	x is not an element of A.	$4 \notin \{1, 2, 3\}$
$A \subseteq B$	A is a subset of B	Every element of A is also an element of B.	$\{1, 2\} \subseteq \{1, 2, 3\}$
$B \supseteq A$	B is a superset of A	B contains every element of A.	$\{1, 2, 3\} \supseteq \{1, 2\}$
$A \subset B$	A is a proper subset of B	$A \subseteq B$ and $A \neq B$.	$\{1, 2\} \subset \{1, 2, 3\}$
$A = B$	A equals B	A and B contain exactly the same elements; order is immaterial.	$\{a, b\} = \{b, a\}$
$A \neq B$	A is not equal to B	The two sets do not contain exactly the same elements.	$\{1, 2\} \neq \{1, 3\}$
$\emptyset$ or $\{ \}$	empty / null set	A set having no element.	$\{x / x \in \mathbb{N}, 1 < x < 2\} = \emptyset$
U or X	universal set	The set containing all sets under consideration as subsets.	$A \subseteq U$
$n(A)$	cardinality of A	Number of distinct elements in finite set A.	$A = \{a, b, c\}; n(A) = 3$
$P(A)$	power set of A	Set of all subsets of A.	$P(\{a\}) = \{\emptyset, \{a\}\}$
A' or A^c	complement of A	Elements of U that are not in A.	$A' = \{x / x \in U, x \notin A\}$

2. Set Operations

Notation	Read as	Definition	Key interpretation
$A \cup B$	A union B	$\{x / x \in A \text{ or } x \in B\}$	All elements in A or B; “or” is inclusive.
$A \cap B$	A intersection B	$\{x / x \in A \text{ and } x \in B\}$	Elements common to both A and B.
$A - B$	A minus B	$\{x / x \in A, x \notin B\}$	Elements in A but not in B.
$B - A$	B minus A	$\{x / x \in B, x \notin A\}$	Elements in B but not in A.
A'	complement of A	$\{x / x \in U, x \notin A\}$	Everything in U outside A.
$A \cap B = \emptyset$	A and B are disjoint	A and B have no common element.	Use $n(A \cup B) = n(A) + n(B)$.
$A \times B$	A cross B	$\{(a, b) / a \in A, b \in B\}$	All ordered pairs with first component from A and second from B.

Useful Region Identities

Identity	Meaning
$A - B = A \cap B'$	In A and outside B.
$B - A = A' \cap B$	Outside A and in B.
$A \cup B = (A - B) \cup (A \cap B) \cup (B - A)$	The union is split into three mutually disjoint regions.
$A = (A \cap B) \cup (A \cap B')$	A is split into the part inside B and the part outside B.
$B = (A \cap B) \cup (A' \cap B)$	B is split into the part inside A and the part outside A.

3. Interval Notation

Interval	Set-builder form	Endpoint rule
(a, b)	$\{x / x \in \mathbb{R}, a < x < b\}$	Excludes both a and b.
$[a, b]$	$\{x / x \in \mathbb{R}, a \leq x \leq b\}$	Includes both a and b.
$[a, b)$	$\{x / x \in \mathbb{R}, a \leq x < b\}$	Includes a; excludes b.
$(a, b]$	$\{x / x \in \mathbb{R}, a < x \leq b\}$	Excludes a; includes b.
(a, ∞)	$\{x / x \in \mathbb{R}, x > a\}$	All real numbers greater than a.
$[a, \infty)$	$\{x / x \in \mathbb{R}, x \geq a\}$	All real numbers greater than or equal to a.
$(-\infty, b)$	$\{x / x \in \mathbb{R}, x < b\}$	All real numbers less than b.
$(-\infty, b]$	$\{x / x \in \mathbb{R}, x \leq b\}$	All real numbers less than or equal to b.
$(-\infty, \infty)$	$\mathbb{R}$	All real numbers. Infinity is never included with a square bracket.

Roster and Set-builder reminders

Rule	What to remember
Roster form	List elements inside braces and separate them by commas. Write repeated elements only once; order is immaterial.
Set-builder form	State the variable, use “/”, and give a property that determines the elements uniquely.
Cardinality	Count distinct elements only. The empty set has cardinality 0 and is a finite set.

4. Laws of Set Algebra

Law / property	Union form	Intersection form / related form
Commutative law	$A \cup B = B \cup A$	$A \cap B = B \cap A$
Associative law	$(A \cup B) \cup C = A \cup (B \cup C)$	$(A \cap B) \cap C = A \cap (B \cap C)$
Identity law	$A \cup \emptyset = A$	$A \cap U = A$
Domination law	$A \cup U = U$	$A \cap \emptyset = \emptyset$
Idempotent law	$A \cup A = A$	$A \cap A = A$
Complement law	$A \cup A' = U$	$A \cap A' = \emptyset$
Double complement	$(A')' = A$	$\emptyset' = U$ and $U' = \emptyset$
Subset result	If $A \subseteq B$, then $A \cup B = B$	If $A \subseteq B$, then $A \cap B = A$
Distributive law	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
De Morgan's laws	$(A \cup B)' = A' \cap B'$	$(A \cap B)' = A' \cup B'$

How to recognise the applicable law

Clue in the expression	Likely tool
A complement outside brackets containing $\cup$ or $\cap$	De Morgan's law: complement each set and interchange $\cup$ with $\cap$.
A set operation repeated with the same set	Idempotent law.
A set together with its complement	Complement law.
$A \cup (B \cap C)$ or $A \cap (B \cup C)$	Distributive law.
One set is known to be a subset of another	Subset union/intersection result.

5. Cardinality Results and Formulae

Situation	Formula / result	Use
Two finite sets	$n(A \cup B) = n(A) + n(B) - n(A \cap B)$	Subtract the overlap counted twice.
Two disjoint sets	$n(A \cup B) = n(A) + n(B)$	Because $n(A \cap B) = 0$.
Complement	$n(A') = n(U) - n(A)$	Count elements outside A but inside U.
Partition of A	$n(A \cap B') + n(A \cap B) = n(A)$	Useful for "A only".
Partition of B	$n(A' \cap B) + n(A \cap B) = n(B)$	Useful for "B only".
Union by three regions	$n(A \cap B') + n(A \cap B) + n(A' \cap B) = n(A \cup B)$	Adds A only, both, and B only.
Three finite sets	$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$	Inclusion-exclusion for three sets.
Power set	If $n(A) = m$, then $n(P(A)) = 2^m$	Number of all subsets of A.
Neither A nor B	$n(A' \cap B') = n(U) - n(A \cup B)$	By De Morgan: $(A \cup B)' = A' \cap B'$.
Exactly one of A and B	$n(A \cap B') + n(A' \cap B) = n(A) + n(B) - 2n(A \cap B)$	Counts A only plus B only.

Standard counting words

Words	Translate to
at least one / A or B	$A \cup B$
both / common	$A \cap B$
neither	$(A \cup B)' = A' \cap B'$
only A	$A - B = A \cap B'$
only B	$B - A = A' \cap B$
all three	$A \cap B \cap C$

6. Ordered Pairs and Cartesian Products

Result	Statement	Practical use
Equality of ordered pairs	$(a,b)=(c,d)$ if and only if $a=c$ and $b=d$.	Equate corresponding components.
Order matters	(a,b) and (b,a) are generally different.	Do not interchange components.
When reversal is equal	$(a,b)=(b,a)$ if and only if $a=b$.	Both components must be equal.
Cartesian product	$A \times B = \{(a,b) / a \in A, b \in B\}$.	First component comes from A; second from B.
Empty factor	If $A=\emptyset$ or $B=\emptyset$, then $A \times B=\emptyset$.	No ordered pair can be formed.
Number of ordered pairs	If $n(A)=m$ and $n(B)=n$, then $n(A \times B)=mn$.	Multiply the two cardinalities.
Order of products	$A \times B \neq B \times A$ in general, but $n(A \times B)=n(B \times A)$.	The pairs change although the counts agree.

Quick construction rule

To list $A \times B$, take each element of A in turn and pair it with every element of B. For $B \times A$, reverse which set supplies the first component.

7. Relations: Symbols, Definitions and Results

Term / notation	Meaning	Result to remember
R from A to B	Any subset of $A \times B$.	$R \subseteq A \times B$.
$(x,y) \in R$ or xRy	x is related to y under R.	x is the pre-image; y is the image.
Domain(R)	Set of all first components in R.	$\{a \in A / (a,b) \in R \text{ for some } b\}$
Co-domain	The target set B when R is from A to B.	The range is always a subset of the co-domain.
Range(R)	Set of all second components occurring in R.	$\{b \in B / (a,b) \in R \text{ for some } a\}$
Binary relation on A	Any subset of $A \times A$.	Domain set and co-domain set are both A.
Empty relation	$R = \emptyset$.	No ordered pair is related.
Universal relation on A	$R = A \times A$.	Every possible ordered pair is related.
Number of relations A to B	If $n(A)=m$ and $n(B)=n$, total relations = 2^{mn} .	There are mn pairs in $A \times B$; a relation is any subset.

Types by mapping behaviour

Type	Test
One-one (injective)	Every element of A has at most one image, and distinct elements of A have distinct images.
Many-one	Two or more elements of A have the same image in B.
Into	At least one element of B has no pre-image; $\text{Range}(R) \subset B$.
Onto (surjective)	Every element of B is an image of some element of A; $\text{Range}(R)=B$.

8. Properties of Relations

Property	Formal test	How to check
Reflexive	$(a,a) \in R$ for every $a \in A$.	Every element must have its self-pair.
Symmetric	$(a,b) \in R \Rightarrow (b,a) \in R$ for all $a,b \in A$.	Reverse every pair; the reversed pair must also belong to R .
Transitive	$(a,b) \in R$ and $(b,c) \in R \Rightarrow (a,c) \in R$.	Whenever two pairs form a chain, the shortcut pair must be present.
Equivalence relation	R is reflexive, symmetric and transitive.	All three tests must be satisfied.

9. Final Problem-solving Checklist

Question type	Minimum facts to write before calculating
Set operation	Write the sets in roster form if useful; identify U before using complements; translate the operation into words.
Venn / survey	Define the sets, write every given cardinality, place the deepest intersection first, and apply inclusion-exclusion.
Power set	List every subset systematically or use 2^m for the count. Remember that $\emptyset$ and A are subsets of A .
Intervals	Check each endpoint separately. Round bracket means excluded; square bracket means included.
Ordered pairs	Equate first with first and second with second.
Cartesian product	Preserve order and ensure every possible pair is included exactly once.
Relation	First verify $R \subseteq A \times B$; then read domain from first components and range from second components.
Relation properties	Test reflexive, symmetric and transitive separately. One counterexample is enough to disprove a property.

Textbook fidelity note

This guide preserves the Unit 1 notation and statements while displaying symbols clearly. It uses “/” in set-builder notation and distinguishes $\in$ (element of) from $\subseteq$ (subset of).